

Olympiad Test Paper — Science (Class 7)

Chapter 17: Forest: Our Lifeline

Subject	Science (NCERT Class 7)
Chapter	17 — Forest: Our Lifeline
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. A forest is a living system — trees, shrubs, herbs, climbers, animals, birds, insects and micro-organisms all linked together. Two short forest food-web descriptions appear in the paper; read them carefully and answer the questions that follow. Most questions need two to four reasoning steps; watch for traps (a forest is not just trees, and decomposers are helpful, not harmful). Do not write on this paper — mark answers on your answer sheet.

1. A forest is best described as:

- (A) a large number of tall trees growing close together
- (B) a place where only wild animals live, with no plants
- (C) any open area covered with grass and a few bushes
- (D) a system of trees, shrubs, herbs, climbers, animals and micro-organisms living together

2. The topmost, leafy "roof" of a forest, formed by the branches of the tallest trees, is called the:

- (A) understorey
- (B) forest floor
- (C) humus layer
- (D) canopy

3. The branchy, leafy top part of a single tree is its:

- (A) crown
- (B) canopy
- (C) understorey
- (D) trunk

4. Small trees and shrubs that grow in the shade below the tallest trees make up the:

- (A) canopy
- (B) understorey
- (C) crown
- (D) emergent layer

5. Plants of the **forest floor**, such as herbs and ferns, usually receive:

- (A) more sunlight than the canopy trees (B) exactly the same sunlight as the canopy
- (C) no light at all, so they make food from soil
- (D) only the sunlight that filters down through the upper layers

6. The branches at the top of tall forest trees interlock to form a canopy. The main reason different plants grow at different heights in a forest is:

- (A) the soil is different at each height (B) competition for animals
- (C) competition for sunlight (D) taller plants are always older

7. A herb that needs full, direct sunlight is most likely to grow well:

- (A) in a sunny gap where a big tree has fallen
- (B) on the dark forest floor under a dense canopy
- (C) only in the topmost canopy (D) inside the trunk of a tree

8. Which set lists the forest layers in order from **top to bottom**?

- (A) forest floor → shrubs → understory → canopy
- (B) canopy → understory → shrubs → forest floor
- (C) shrubs → canopy → forest floor → understory
- (D) understory → canopy → shrubs → forest floor

9. On the forest floor, dead leaves, twigs and animal droppings are broken down mainly by:

- (A) tall canopy trees (B) sunlight alone
- (C) decomposers such as fungi and micro-organisms
- (D) grazing deer

10. The dark, soft, nutrient-rich material formed when decomposers break down dead leaves and droppings is called:

- (A) sand (B) canopy
- (C) humus (D) nectar

11. A student says, "Decomposers are harmful because they make things rot and should be removed from the forest." The best correction is:

- (A) correct — rotting spreads disease, so decomposers are harmful
- (B) decomposers are essential; they recycle dead matter into humus that feeds new plants
- (C) decomposers only live in ponds, not forests
- (D) decomposers eat living trees, so they damage the forest

12. Imagine every decomposer in a forest suddenly vanished. Over time the forest would most likely:

- (A) stay exactly the same
- (B) become buried under dead matter and starved of recycled nutrients

- (C) grow faster because nothing breaks down its leaves
- (D) immediately turn into a desert overnight

13. Decomposers return the nutrients from dead plants and animals to the:

- (A) air, where plants cannot use them
- (B) canopy, where animals eat them
- (C) soil, where plant roots can take them up again
- (D) sunlight, which carries them away

14. The continuous movement of nutrients from soil → plants → animals → dead matter → decomposers → soil is best called the:

- (A) water cycle only
- (B) nutrient (recycling) cycle of the forest
- (C) food chain reversing itself
- (D) carbon-only cycle

15. In a forest food chain, the **producers** are the:

- (A) tigers and other carnivores
- (B) deer and other herbivores
- (C) green plants
- (D) fungi and bacteria

16. In the food chain grass → deer → tiger, the deer is a:

- (A) producer
- (B) decomposer
- (C) herbivore (and the tiger's prey)
- (D) carnivore that eats the tiger

17. A **food web** differs from a single food chain because a food web:

- (A) has no producers
- (B) leaves out all the animals
- (C) always has exactly three organisms
- (D) shows many interlinked food chains in the forest

> **Forest food web (for Q18–Q22).** In one forest: grass and leaves are eaten by grasshoppers and by deer. Grasshoppers are eaten by frogs; frogs are eaten by snakes; snakes are eaten by hawks. Deer are eaten by tigers. Hawks also catch and eat small birds, and those small birds eat grasshoppers too.

18. In this web, the grasshopper is eaten by both:

- (A) grass and frogs
- (B) tigers and hawks
- (C) deer and snakes
- (D) frogs and small birds

19. If a disease killed almost all the **frogs** in this web, the most direct effect would be:

- (A) the grass would disappear
- (B) grasshopper numbers would tend to rise and snake food would fall
- (C) tigers would immediately die out
- (D) nothing would change anywhere

20. In this web, which animal eats prey from **two different** food chains?

- (A) the hawk (it eats both snakes and small birds)
- (B) the grasshopper
- (C) the deer
- (D) the frog

- 21.** The ultimate source of energy for **every** organism in this forest web is:
(A) the tiger at the top (B) the humus in the soil
(C) the Sun, captured by green plants (D) the decomposers on the floor
- 22.** If hunters removed all the **tigers** from this web, the most likely first effect is that:
(A) the deer population would tend to increase (B) the grass would vanish at once
(C) the hawks would stop eating snakes (D) the frogs would start eating tigers
- 23.** Forests are often called the "green lungs" of the Earth because their plants:
(A) take in oxygen and release carbon dioxide
(B) release oxygen and absorb carbon dioxide (C) breathe exactly like animal lungs
(D) store water in the air
- 24.** During the day, a forest helps keep the air's gases balanced mainly because its plants:
(A) only respire and give out CO₂
(B) photosynthesise, taking in CO₂ and giving out O₂
(C) absorb all oxygen from the air (D) make no gases at all
- 25.** Cutting down a large area of forest tends to **increase** the amount of which gas in the air?
(A) oxygen (B) water vapour only
(C) nitrogen only (D) carbon dioxide
- 26.** Forest roots help **prevent floods and soil erosion** mainly by:
(A) gripping the soil and slowing rainwater so it soaks in
(B) drinking up entire rivers (C) blocking sunlight from the ground
(D) pushing the soil downhill faster
- 27.** After a hillside forest is cut down, heavy rain on the bare slope is most likely to cause:
(A) less soil erosion than before
(B) more soil erosion and a greater chance of floods
(C) the soil to become richer in humus (D) no change to the slope at all
- 28.** Forests help **recharge groundwater** because tree cover and leaf litter:
(A) turn groundwater into oxygen (B) seal the soil so no water can enter
(C) send all rainwater straight to the sea
(D) let rainwater soak slowly into the ground instead of running off
- 29.** Through transpiration and shade, large forests tend to influence the local climate by:
(A) making the area hotter and drier (B) stopping all wind permanently
(C) adding moisture to the air and helping bring rainfall
(D) having no effect on rainfall at all
- 30.** A forest is called a "lifeline" because it provides, all at once:

- (A) only timber for furniture
- (B) oxygen, clean water, soil protection, climate control and habitat for wildlife
- (C) only a home for tigers
- (D) only firewood for villages

31. Which of these is a **product** we get from forests?

- (A) plastic
- (B) petrol
- (C) gum, honey, timber and medicinal herbs
- (D) glass

32. A single fallen forest tree slowly rots over years. The nutrients locked in its wood are:

- (A) lost forever from the forest
- (B) returned to the soil by decomposers for new plants to use
- (C) carried up into the canopy
- (D) turned directly into sunlight

33. Which chain correctly shows the path of **energy** in a forest?

- (A) tiger → deer → grass → Sun
- (B) deer → tiger → grass → Sun
- (C) grass → Sun → tiger → deer
- (D) Sun → grass → deer → tiger

34. In a forest, the number of **carnivores** like tigers is usually much smaller than the number of deer because:

- (A) much less energy is available higher up the food chain
- (B) tigers cannot move
- (C) deer never get eaten
- (D) carnivores do not need food

35. A wheat field has millions of plants of one kind growing together for one season. It is **not** called a forest because a forest:

- (A) must contain at least one tiger
- (B) is a stable, many-layered community of many different species lasting many years
- (C) is always smaller than a field
- (D) has no animals in it

36. Which statement about a forest is **correct**?

- (A) a forest is simply a collection of trees and nothing else
- (B) decomposers play no role in a forest
- (C) forests have no effect on the air we breathe
- (D) a forest is a community of plants, animals and micro-organisms that depend on one another

37. After a forest fire clears an open patch, over the following years the area usually:

- (A) stays bare rock forever
- (B) becomes a desert that never recovers
- (C) turns instantly into a full grown forest the next morning
- (D) is recolonised by herbs, then shrubs, then trees — the forest regrows

38. The fact that a burnt or cleared forest can slowly grow back shows that a forest is:

- (A) dynamic — it grows and regenerates naturally over time
- (B) completely unchanging and fixed
- (C) man-made and cannot recover on its own
- (D) the same as a crop field

39. Of canopy, understorey and forest floor, the layer that receives the **least** sunlight is the:

- (A) forest floor
- (B) understorey
- (C) canopy
- (D) all three receive equal sunlight

40. Which organism in a forest is a **decomposer**?

- (A) a deer
- (B) a mango tree
- (C) a mushroom (fungus)
- (D) a hawk

> **Forest food web (for Q41–Q45).** In a second forest: trees and shrubs are eaten by insects and by rabbits. Insects are eaten by small birds; small birds are eaten by snakes. Rabbits are eaten by foxes. Snakes are also eaten by foxes. On the floor, fungi and bacteria break down everything that dies.

41. In this second web, the fox gets its food from:

- (A) plants directly
- (B) two chains — it eats both rabbits and snakes
- (C) insects only
- (D) the fungi on the floor

42. If the **rabbits** in this web were wiped out, which animal would be most directly short of food?

- (A) the snake
- (B) the small bird
- (C) the insect
- (D) the fox (it loses one of its prey)

43. In this second web, the dead bodies of foxes, snakes and trees are returned to the soil by the:

- (A) fungi and bacteria (decomposers)
- (B) small birds
- (C) rabbits
- (D) insects

44. If the number of **insects** in this web increased sharply, the most likely first effect is that:

- (A) small birds would have more food and could increase
- (B) the trees would grow taller
- (C) the foxes would disappear at once
- (D) the fungi would stop working

45. Counting the chains in this web, the rabbit and the insect both depend, in the end, on the:

- (A) foxes for energy
- (B) snakes for energy
- (C) trees and shrubs (the producers) for energy
- (D) decomposers for energy

46. Match the layer to its typical resident: the **canopy** is the most likely home of:

- (A) birds and monkeys feeding among the high branches
- (B) ground beetles in the litter
- (C) earthworms and fungi
- (D) moss growing in deep shade

47. Earthworms, fungi and many beetles live mainly on or in the:
- (A) canopy (B) crowns of the tallest trees
(C) emergent layer (D) forest floor
48. Which is the best reason a forest's soil is usually **rich and fertile**?
- (A) the canopy blocks all the rain (B) animals plough the soil with their feet
(C) decomposers constantly turn fallen leaves and dead matter into humus
(D) there is no sunlight at the bottom
49. A keystone animal such as the tiger keeps deer numbers in check. If tigers vanish, deer may multiply and overgraze. This shows that in a forest:
- (A) removing one species can disturb many others
(B) animals have no effect on plants
(C) plants and animals are completely independent
(D) more deer is always good for the forest
50. Which chain of events follows large-scale **deforestation**?
- (A) fewer trees → more soil erosion, less rainfall, lost habitat and more CO₂
(B) more trees → more rain → richer soil (C) fewer trees → more oxygen in the air
(D) fewer trees → groundwater rises everywhere
51. A villager notices the stream near a freshly cleared forest now floods in the rains and dries up in summer. The best explanation is that, without the forest:
- (A) the trees were drinking all the floodwater (B) the Sun stopped shining on the hill
(C) the stream moved to a new place
(D) rainwater now runs off fast instead of soaking in to feed the stream slowly
52. The plants of a forest are the **producers** because they:
- (A) eat other animals (B) break down dead leaves
(C) make their own food using sunlight, water and carbon dioxide
(D) cannot make food and must be fed
53. Which pairing of forest role and organism is **correct**?
- (A) producer — tiger (B) herbivore — deer
(C) decomposer — deer (D) carnivore — grass
54. Two students argue. Aman: "A forest is only useful for the wood we cut from it." Bina: "A forest gives oxygen, holds the soil, recharges water and shelters wildlife." Who is right?
- (A) Aman, because wood is the only product (B) both are equally wrong
(C) Bina, because a forest provides many life-supporting services
(D) neither — forests do nothing useful
55. In the food chain leaves → caterpillar → small bird → snake, the **snake** is a:

- (A) carnivore at the top of this chain (B) herbivore
(C) producer (D) decomposer

56. Why does removing the **canopy** trees badly hurt the lower forest layers and animals?

- (A) it suddenly exposes shade plants and floor animals to harsh sun, heat and rain
(B) it has no effect on them (C) it makes the floor colder than the canopy
(D) it gives the floor plants too little carbon dioxide

57. Sunlight filtering through the canopy is **important** for the forest because it:

- (A) is harmful and should be blocked completely
(B) lets shade-loving understory and floor plants make enough food to survive
(C) only warms the animals, not the plants (D) has no role once the canopy exists

58. A forest captures and stores carbon mainly in its:

- (A) the empty air between trees (B) animals' fur
(C) flowing streams
(D) wood, leaves and soil (as plants take in CO₂)

59. Putting it together: which single statement best explains why a forest is called "our lifeline"?

- (A) it only gives us timber and firewood to sell
(B) it matters only to the animals that live in it, not to people
(C) it is simply a large group of trees with no other use
(D) it supplies oxygen, recycles nutrients, holds soil and water, balances climate, and shelters countless species — all together

60. Which action would best help **conserve** a forest and the services it provides?

- (A) clearing it for a single crop to use the land
(B) removing all the decomposers to stop rotting
(C) protecting it and planting trees where forest has been lost
(D) cutting the canopy so more light reaches the floor

End of question paper. Maximum marks: 240. Marking: +4 correct, -1 wrong, 0 unattempted.

Solutions — Science (Class 7), Chapter 17: Forest: Our Lifeline

Marking: +4 correct, –1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	11	B	21	C	31	C	41	B	51	D
2	D	12	B	22	A	32	B	42	D	52	C
3	A	13	C	23	B	33	D	43	A	53	B
4	B	14	B	24	B	34	A	44	A	54	C
5	D	15	C	25	D	35	B	45	C	55	A
6	C	16	C	26	A	36	D	46	A	56	A
7	A	17	D	27	B	37	D	47	D	57	B
8	B	18	D	28	D	38	A	48	C	58	D
9	C	19	B	29	C	39	A	49	A	59	D
10	C	20	A	30	B	40	C	50	A	60	C

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

- (D)** A forest is not just a crowd of tall trees — it is a whole system of trees, shrubs, herbs, climbers, animals, birds, insects and micro-organisms living together and depending on one another. The other options leave out most of the community.
- (D)** The canopy is the topmost branchy "roof" made by the crowns of the tallest trees. The understorey and forest floor are lower layers; humus is decomposed matter in the soil.
- (A)** The crown is the branchy, leafy top of a single tree. The canopy is the combined roof formed by many crowns together — a property of the whole forest, not one tree.
- (B)** Smaller trees and shrubs growing in the shade beneath the tallest trees form the understorey. The canopy and emergent layer are at the top; the crown belongs to one tree.
- (D)** Floor herbs and ferns get only the sunlight that filters down through the canopy and understorey above them. They still make their own food by photosynthesis (so "from soil" is wrong); they certainly do not get more light than the canopy.

- 6. (C)** Plants grow at different heights chiefly because they compete for sunlight — the tallest grab it first, and shorter plants survive on what filters through. Soil and age are not the main driver.
- 7. (A)** A sun-loving herb does best in a bright gap where a big tree has fallen and light reaches the ground. Under a dense canopy it is too shady; it cannot live up in the canopy or inside a trunk.
- 8. (B)** From top to bottom the layers run canopy → understory → shrubs → forest floor. The other orders jumble these levels.
- 9. (C)** Dead leaves, twigs and droppings on the floor are broken down by decomposers — fungi and micro-organisms. Canopy trees, sunlight alone and grazing deer do not do this recycling work.
- 10. (C)** The dark, nutrient-rich material made when decomposers break down dead matter is humus. Canopy is a layer, sand is mineral grains, and nectar is a flower sugar.
- 11. (B)** This is the decomposer trap. Decomposers are essential, not harmful: by rotting dead matter into humus they recycle nutrients that feed new plants. Without them the forest would choke on dead material.
- 12. (B)** With no decomposers, dead leaves and bodies would pile up and the locked-in nutrients would never return to the soil — the forest would be buried in dead matter and starved of recycled nutrients. (It would not improve, nor change overnight.)
- 13. (C)** Decomposers return nutrients to the soil, where plant roots can absorb them again. This closes the loop; the air, canopy and sunlight do not store these nutrients for roots.
- 14. (B)** The repeating path soil → plants → animals → dead matter → decomposers → soil is the forest's nutrient (recycling) cycle. It involves more than water and more than carbon, and it is not a food chain run backwards.
- 15. (C)** Producers are the green plants — they make food from sunlight. Herbivores (deer), carnivores (tiger) and decomposers (fungi) all depend on that food; they do not produce it.
- 16. (C)** In grass → deer → tiger, the deer eats the plant (so it is a herbivore) and is itself eaten by the tiger (so it is the tiger's prey). It is neither a producer nor a carnivore.
- 17. (D)** A food web is many food chains interlinked, because most animals eat, and are eaten by, more than one kind of organism. It still has producers and animals; it is not fixed at three organisms.
- 18. (D)** Trace Web 1: grasshoppers are eaten by frogs, and the small birds also eat grasshoppers. So the grasshopper is eaten by both frogs and small birds. Tigers eat deer

(not grasshoppers); snakes eat frogs.

19. (B) Frogs eat grasshoppers and are eaten by snakes. Remove most frogs and grasshoppers face one fewer predator (numbers tend to rise) while snakes lose a food source (their food falls). Grass would not vanish, and tigers (on the deer chain) are unaffected at first.

20. (A) The hawk eats snakes (grass → grasshopper → frog → snake → hawk) and also eats small birds (grass → grasshopper → small bird → hawk) — prey from two different chains. The deer, grasshopper and frog each sit on a single chain here.

21. (C) Every chain in the web starts with green plants, which capture energy from the Sun. So the ultimate energy source for all organisms is the Sun. The tiger, humus and decomposers are not energy sources for the whole web.

22. (A) Tigers eat deer. Remove the tigers and the deer lose their main predator, so the deer population tends to increase first. Grass would not vanish at once, and the hawk-snake link is on another chain.

23. (B) Forests are the "green lungs" because their plants release oxygen and absorb carbon dioxide during photosynthesis. They do not breathe like animal lungs, and option A reverses the gases.

24. (B) By day, forest plants photosynthesise — taking in CO₂ and giving out O₂ — which helps keep the air's gases in balance. They do not only respire, nor absorb all the oxygen, nor make no gases.

25. (D) Cutting a large forest removes plants that were absorbing CO₂, so the amount of carbon dioxide in the air tends to rise. Oxygen, nitrogen and water vapour are not the gas that increases here.

26. (A) Roots grip the soil and the leaf litter slows the rain so water soaks in rather than rushing off — this is how forests curb erosion and floods. They do not drink rivers dry or push soil downhill.

27. (B) Strip a hillside of forest and there are no roots to hold the soil or slow the rain, so heavy rain causes more soil erosion and a greater flood risk — the opposite of "less erosion."

28. (D) Forest cover and litter let rainwater soak slowly into the ground, recharging the groundwater, instead of running straight off. They do not seal the soil or rush water to the sea.

29. (C) Through transpiration (releasing water vapour) and shade, large forests add moisture to the air and help bring rainfall, cooling and dampening the local climate — they

do not dry the area or stop wind.

30. (B) A forest is a "lifeline" because it gives oxygen, clean water, soil protection, climate control and wildlife habitat — all at once. The other options each name just one narrow use.

31. (C) Gum, honey, timber and medicinal herbs are genuine forest products. Plastic, petrol and glass are manufactured from other raw materials, not harvested from a forest.

32. (B) As a fallen tree rots, decomposers return the nutrients in its wood to the soil for new plants to use — they are recycled, not lost, raised to the canopy, or turned into sunlight.

33. (D) Energy flows Sun → grass → deer → tiger: plants capture the Sun's energy, herbivores eat plants, carnivores eat herbivores. The other chains run the arrows backwards or jumble the order.

34. (A) Much less energy is available higher up a food chain (most is used up at each step), so the top carnivores like tigers can only be few, while the deer below them are many. The other options are false.

35. (B) A wheat field is one species, one layer, lasting one season; a forest is a stable, many-layered community of many different species lasting many years — that is the real difference, not the presence of a tiger or its size.

36. (D) The correct statement is that a forest is a community of plants, animals and micro-organisms that depend on one another. "Just trees," "no effect on air," and "decomposers play no role" are all wrong.

37. (D) After a clearing, the patch is recolonised step by step — first herbs, then shrubs, then trees — and the forest regrows. It neither stays bare forever nor springs up overnight nor becomes permanent desert.

38. (A) The fact that a burnt or cleared forest grows back shows a forest is dynamic — it grows and regenerates naturally over time. It is therefore not fixed, not man-made-only, and not the same as a crop field.

39. (A) Sunlight is caught first by the canopy, less reaches the understorey, and least of all reaches the forest floor at the bottom. So the forest floor receives the least sunlight.

40. (C) A mushroom is a fungus — a decomposer that breaks down dead matter. A deer is a herbivore, a mango tree a producer, and a hawk a carnivore.

41. (B) In Web 2 the fox eats rabbits (trees → rabbit → fox) and also eats snakes (trees → insect → small bird → snake → fox) — food from two chains. It does not eat plants, insects directly, or the floor fungi.

- 42. (D)** Foxes eat both rabbits and snakes. Wipe out the rabbits and the fox loses one of its two prey, so it is the animal most directly short of food. Snakes, small birds and insects do not feed on rabbits.
- 43. (A)** When foxes, snakes and trees die, the fungi and bacteria (decomposers) on the floor break them down and return their nutrients to the soil. Rabbits, small birds and insects are not decomposers.
- 44. (A)** Small birds eat insects, so a sharp rise in insects means more food for small birds, whose numbers could then increase first. The trees, foxes and fungi are not the most direct first responders.
- 45. (C)** Follow every chain back to its start: both the rabbit (trees → rabbit) and the insect (trees → insect) feed on the trees and shrubs — the producers — for their energy. Foxes, snakes and decomposers are not the base.
- 46. (A)** The canopy, high among the branches, is the typical home of birds and monkeys. Earthworms, fungi, litter beetles and deep-shade moss all belong to the forest floor, not the canopy.
- 47. (D)** Earthworms, fungi and many beetles live mainly on or in the forest floor, among the leaf litter — not in the canopy, crowns or emergent layer at the top.
- 48. (C)** Forest soil is rich because decomposers constantly turn fallen leaves and dead matter into humus, recharging the soil with nutrients. Blocked rain, animal feet and lack of sunlight do not make soil fertile.
- 49. (A)** When losing the tiger lets the deer multiply and overgraze, it shows that removing one species can disturb many others in a forest. Animals clearly do affect plants, and more deer is not always good.
- 50. (A)** Large-scale deforestation sets off: fewer trees → more soil erosion, less rainfall, lost habitat and more CO₂. The other chains wrongly claim benefits from cutting trees.
- 51. (D)** Without the forest, rainwater now runs off fast instead of soaking in, so the stream floods in the rains and dries up in summer — the slow, steady release is gone. The trees were not "drinking floodwater," and the Sun and stream did not move.
- 52. (C)** Plants are producers because they make their own food using sunlight, water and carbon dioxide. They do not eat animals, break down dead leaves, or depend on being fed.
- 53. (B)** The correct pairing is herbivore — deer. Tigers are carnivores (not producers), deer are not decomposers, and grass is a producer (not a carnivore).
- 54. (C)** Bina is right: a forest provides many life-supporting services — oxygen, soil holding, water recharge and wildlife shelter — not just wood. Aman's "wood only" view

misses almost everything a forest does.

55. (A) In leaves → caterpillar → small bird → snake, the snake eats the bird and nothing in this chain eats the snake, so it is a carnivore at the top of this chain — not a producer, herbivore or decomposer.

56. (A) Removing the canopy suddenly exposes shade-adapted plants and floor animals to harsh sun, heat and pounding rain that they are not built for, harming the lower layers. It does not leave them unaffected or make the floor colder.

57. (B) Filtered sunlight matters because it lets shade-loving understorey and floor plants make enough food to survive. The light is helpful, not harmful, and it feeds plants, not just animals.

58. (D) A forest stores carbon mainly in its wood, leaves and soil as the plants take in CO₂ and build it into their tissues and humus. Animal fur, streams and the air between trees are not the main carbon stores.

59. (D) A forest is "our lifeline" because, all together, it supplies oxygen, recycles nutrients, holds soil and water, balances the climate and shelters countless species. The single-use options each miss the bigger picture.

60. (C) Conserving a forest means protecting it and planting trees where forest has been lost (afforestation). Clearing it for one crop, removing decomposers, or cutting the canopy would all damage the forest, not conserve it.